Fast Neural Style Transfer

Efficient Artistic Transformation using VGG19

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Problem Statement

Defining the boundaries and constraints of artistic synthesis.

Input, Output & Constraints

System Dynamics

Input: A *Content Image* (providing semantic structure) and a *Style Image* (providing textures/patterns).

Output: A synthesized image that blends the identity of the content with the aesthetic of the style.

Primary Constraints

Performance: Standard NST is iterative and slow (tens of seconds).

Scalability: High resolution and deep networks lead to significant VRAM consumption and computational latency.

Literature Review

Evolution from classical methods to fast optimizations.

Classical NST vs. Fast Method

The Baseline (Gatys et al.)

Relies on iterative optimization over deep feature spaces in VGG19. It extracts content from deep layers and style via Gram Matrices across multiple layers.

Result: High stylistic richness but extremely high computational cost.

How Our Work Differs

Instead of relying on standard depth, we propose a **Simplified Variant**. By limiting feature complexity and reducing iterations, we optimize for speed without losing semantic integrity.

Methodology

Proposed simplification steps for computational efficiency.

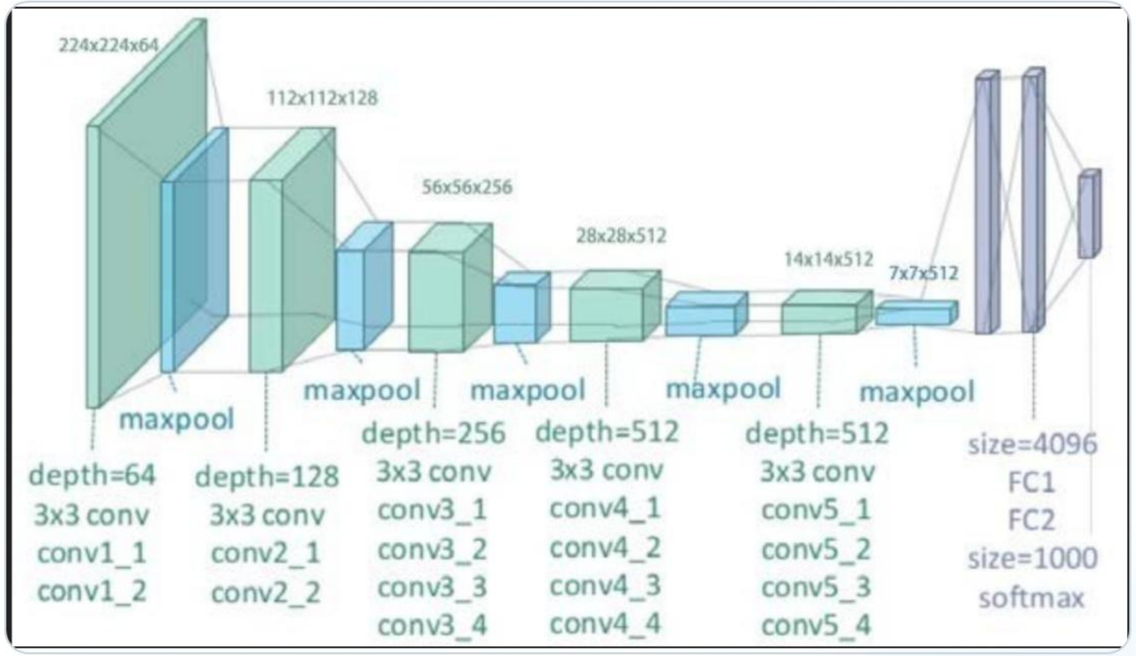
Proposed Method: System Overview

Architecture Design

We use a pre-trained VGG19 (frozen weights) as a feature extractor. The content loss measures structural similarity, while the style loss uses Gram matrices to match patterns.

$$L_{total} = \alpha L_{content} + \beta L_{style}$$

Innovation: A user-tunable α parameter allows dynamic control over stylization strength.



Specific Simplification Steps

- **Resolution Reduction:** Scaled down from 512x512 to 256x256, reducing pixel-wise calculations by 4x.
- **Pruned Layer Depth:** Reduced style layers to conv_1 and conv_2 only, focusing on primary textures rather than abstract deep features.
- **Iteration Optimization:** Iteration count decreased from 300 to 100, accelerating convergence significantly.
- **Hardware Efficiency:** Lower complexity makes the model suitable for low-power or real-time environments.

Implementation Configuration

Parameter	Baseline (Full Model)	Simplified (Fast Model)
Resolution	512 x 512	256 x 256
Style Layers	conv_1 to conv_5	conv_1, conv_2
Content Layer	conv_4	conv_2
Iterations	300	100

Visual Comparison: Case Studies

Target Style (Starry Night)

Case 1: Content (House)

Case 2: Content (Jet)

The Fast model preserves structural content (House/Jet) while applying a simplified version of the Starry Night aesthetic.

Quantitative Performance (Runtime)

Net Speedup: ~10.75x to 11.99x

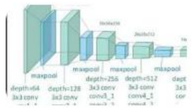
Conclusion & Novelty

Our approach achieves an average **12x speedup** through resolution and network depth pruning.

Novelty: Dynamic trade-off balancing content preservation and computational latency for real-time applications.

Questions?

Image Sources



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Source: medium.com

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https://upload.wikimedia.org/wikipedia/commons/thumb/e/ea/Van_Gogh_-_Starry_Night_-_Google_Art_Project.jpg/3840px-Van_Gogh_-_Starry_Night_-_Google_Art_Project.jpg

Source: commons.wikimedia.org

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https://d3rcx32iafnn0o.cloudfront.net/Pictures/480x320fitpad[0]/9/9/5/2003995_vestige_danielhopkinson_original_5_689009.jpg

Source: www.bdonline.co.uk

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https://static.vecteezy.com/system/resources/previews/065/635/805/large_2x/military-jet-flying-above-fluffy-clouds-in-a-clear-blue-sky-photo.jpg

Source: www.vecteezy.com